

One-Copy-Per-Sector Scope Restriction in the Fundamental Theorem Formalization

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1 The source argument

Cirac–Pérez-García–Schuch–Verstraete 2016 states the general proportional Fundamental Theorem as Corollary `II_cor2` of [CPGSV16]. The theorem applies to two tensors A and B whose matrix-product vectors are proportional, each presented in its basis-of-normal-tensor (BNT) canonical form. The canonical form for A is a direct-sum decomposition

$$A^i = \bigoplus_{j=1}^{g_A} I_{r_{A,j}} \otimes \mu_{A,j} (A_{j,1}^i \oplus \cdots \oplus A_{j,r_{A,j}}^i),$$

where each modulus class j contains $r_{A,j} \geq 1$ representative blocks $A_{j,1}, \dots, A_{j,r_{A,j}}$ all sharing the same weight modulus $|\mu_{A,j}|$, and the moduli are strictly ordered across classes: $|\mu_{A,1}| > |\mu_{A,2}| > \cdots > |\mu_{A,g_A}|$. The same structure applies to B with multiplicity counts $r_{B,j}$.

The proof of Corollary `II_cor2` in [CPGSV16] proceeds in several steps. Step 3 (lines 1156–1163 of the arXiv version) derives the blockwise power-sum identity

$$\sum_{q=1}^{r_{A,j}} \mu_{A,j,q}^N = \sum_{q=1}^{r_{B,j}} (\nu_{B,j,q} e^{i\phi_j})^N, \quad \text{for all } j \text{ and all } N \geq 1, \quad (1)$$

from the BNT linear-independence input together with the equal-MPV hypothesis (so the equality of matrix-product vectors forces the equality of the sums of N th powers). Step 4 invokes Lemma `Lem:app_simple` of the same paper: if two finite sequences $(\lambda_{a,k})_{k=1}^{x_a}$ and $(\lambda_{b,k})_{k=1}^{x_b}$ satisfy

$$\sum_{k=1}^{x_a} \lambda_{a,k}^N = \sum_{k=1}^{x_b} \lambda_{b,k}^N \quad \text{for } N = 1, \dots, \max(x_a, x_b),$$

then $x_a = x_b$ and the two sequences are equal as multisets. Applied to (1), this recovers $r_{A,j} = r_{B,j} =: r_j$ and the multiset equality $\{\mu_{A,j,q}\} = \{e^{i\phi_j} \nu_{B,j,q}\}$ for each modulus class j . Step 6 then assembles the global gauge isomorphism

as $Y = \bigoplus_j I_{r_j} \otimes Y_j$, using the per-class dimension matching $r_{A,j} = r_{B,j}$ from Step 4.

The mathematical structure of this argument is that the multiplicities r_j and the weight multisets are recovered from a finite family of polynomial identities in N , without any passage to a limit.

2 The one-copy-per-sector restriction

The surviving formal one-copy canonical form structure¹ imposes *strict* decreasing moduli across the entire index set: every block has a distinct modulus. This is a stronger condition than the source structure, which only requires strict ordering of the modulus *classes* while allowing $r_j \geq 1$ representatives within each class. The effect is to force $r_j = 1$ for every modulus class j : the formal canonical forms are restricted to the one-copy-per-sector special case.

In this special case, the source decomposition reduces to $A^i = \bigoplus_{j=1}^g \mu_j A_j^i$ with a single block per weight, and the general multiplicity structure $(r_{j,1}, \dots, r_{j,r_j})$ collapses to a single representative. The identity (1) becomes the single-term equality $\mu_{A,j}^N = (e^{i\phi_j} \nu_{B,j})^N$, Steps 3–4 of the source proof become vacuous (there is nothing to separate), and the global gauge in Step 6 takes the simpler form $Y = \bigoplus_j Y_j$ (no identity tensor factor). The formalization thus covers only the one-copy-per-sector instance of Corollary II_cor2.

3 Missing components for the general case

Four items are currently absent from the formalization; each is required to lift the scope from $r_j = 1$ to $r_j \geq 1$.

Lemma Lem:app_simple (finite Newton–Girard uniqueness). The source paper states, in its appendix, that two finite sequences whose power sums agree for $N = 1, \dots, \max(x_a, x_b)$ must be equal as multisets. This is a finite algebraic fact, provable by Newton–Girard identities: the power sums determine the elementary symmetric polynomials, which determine the multiset of roots. A module named `ScalarPowerSumIdentity`² was added at an early stage of the project, but it does not specialize to this particular fixed-length statement in the form needed by Step 4 of the source proof. The lemma needs to be formalized as a standalone result, separate from any BNT context.

BNT power-sum identity. The identity (1) requires a proof: starting from the BNT structure on A and B and the equal-MPV hypothesis, one must derive, for each modulus class j , the equality of the two power sums in (1). The formal development no longer packages coefficient arrays for the discarded proportional branch as standalone hypotheses. The source derivation of (1)

¹`TNLean/MPS/BNT/Construction.lean`: `IsNormalCanonicalFormBNT`. The earlier non-normal wrapper was retired after its downstream consumers were replaced by explicit separated-block hypotheses.

²Introduced in commit `dd54b79e`.

from the BNT linear-independence data and the equal-MPV condition is still not carried out. This derivation is logically prior to applying `Lem:app_simple`.

Multi-copy sector structure. The surviving one-copy structure `IsNormalCanonicalFormBNT`, and any future non-normal analogue stated with explicit separated-block hypotheses, would need a field allowing equal-modulus groups: rather than a global strict ordering on the full index set, the condition should be a strict ordering on the set of modulus classes, with each class admitting an arbitrary finite number of representative blocks. Relaxing `mu_strict_anti` to this weaker form is the necessary type-level change. Without it, the general-multiplicity versions of Corollary `II_cor2` and of the gauge assembly step cannot be stated.

Global gauge assembly. With $r_j \geq 1$ copies per sector, the global gauge isomorphism takes the form $Y = \bigoplus_j I_{r_j} \otimes Y_j$, where I_{r_j} is the identity on the multiplicity space and Y_j is the per-block gauge on the bond dimension. Constructing and verifying this more general gauge requires the dimension match $r_{A,j} = r_{B,j}$ from Step 4, which in turn requires `Lem:app_simple` and the power-sum identity. The simpler $Y = \bigoplus_j Y_j$ gauge used in the one-copy case is already present.

4 What the current formalization does prove

Within the one-copy-per-sector restriction, the formalization establishes the equal-MPV branch of the Fundamental Theorem argument. The BNT separation hypotheses yield cross-block overlap decay, and the equal-MPV hypothesis at a single witness length is used to match the per-block gauge. The former one-shot proportional branch has been deleted: the surviving block-matching boundary uses the explicit conclusion `BlockPermutationGaugePhaseConclusion` only where the BNT comparison has already been supplied. Thus no current theorem is presented as deriving the source proportional comparison from the discarded coefficient hypotheses.

The source normalization $|\mu_1| = 1$ is now recorded as the field `mu_dom_norm_one` in `IsNormalCanonicalFormBNT`,³ which brings the dominant-block normalization into alignment with the source convention. Under this normalization and the strict modulus ordering, every sub-dominant block has $|\mu_j| < 1$.

5 Downstream audit status

The downstream CPSV16 surfaces that currently consume the restricted normal-CF-BNT predicate have been audited against the one-copy-per-sector boundary.⁴ Each of these public surfaces either assumes or constructs an

³`IsNormalCanonicalFormBNT.mu_dom_norm_one`, `TNLean/MPS/BNT/Construction.lean` lines 192–198.

⁴The relevant Lean surfaces are `IsNormalCanonicalFormBNT`, `IsNormalCanonicalFormBNT.ofSeparatedData`, `IsNormalCanonicalFormBNT.cross_overlap_tendsto_zero`, `IsNormalCanonicalFormBNT.isBNT`, `IsNormalCanonicalFormBNT.exists_common_blockTensor_isInjective`,

already chosen representative family, strictly ordered by weight modulus. They do not introduce a source-faithful canonical-form existence theorem and do not recover the repeated-copy sector data of the source BNT decomposition. Their docstrings and the corresponding Wielandt blueprint corollary therefore carry the same one-copy-per-sector marker.

This audit does not apply that marker to the sector-BNT development based on `lsBNTCanonicalForm`. That predicate is the paper-faithful sector surface used for the CPSV16 coefficient and weight-multiplicity arguments, where repeated copies inside one sector remain explicit.

6 Conclusion

The present formalization proves the proportional Fundamental Theorem in the one-copy-per-sector special case ($r_j = 1$ for every modulus class j). The general case of Corollary `II_cor2` in [CPGSV16], which allows $r_j \geq 1$, requires three additional components not yet in the formalization: the standalone Newton–Girard uniqueness lemma (`Lem:app_simple`), the derivation of the blockwise power-sum identity from the BNT structure and the equal-MPV hypothesis, and the global gauge assembly with multiplicity identity factors. Supplying these components would also require relaxing the `mu'strict'anti` field from a global strict ordering to a class-level ordering.

The corresponding tracked items are issue #1560 for the standalone `Lem:app_simple` lemma, issue #1561 for the BNT power-sum identity derivation, and issue #1562 for the general multi-copy route tracking the full relaxation.

References

- [CPGSV16] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product density operators: Renormalization fixed points and boundary theories. Preprint; published version: *Annals of Physics* 378 (2017), 100–149, 2016.

and `exists'common'blockTensor'isInjective'two'of'isNormalCanonicalFormBNT`.